

1.1 COURSE SYLLABI

Introduction to Software Technology / Computer Science Principles (Dual Credit)

Pathway: Computer Science Pathway

Course Description:

This yearlong course introduces students to computing, programming, and digital innovation. During Semester 1, students complete Introduction to Software Technology coursework focused on digital literacy, technology systems, web development, robotics, and programming foundations. During Semester 2, students transition into Computer Science Principles concepts through creative computing, programming, data analysis, cybersecurity, ethics, and computing impacts. Students utilize CodeHS, GitHub.dev, Micro:bit devices, and Finch robots while developing employability and technical skills necessary for future careers in technology.

System Grading Scale

- A = 90-100
- B = 80-89
- C = 70-79
- F = Below 70

Class Grading Scale

- Major Assessments = 55%
- Minor Assessments = 35%
- Practice Assignments = 10%

Career Opportunities

- Software Developer
- Web Developer

- Data Analyst
- Cybersecurity Specialist
- Computer Programmer
- Systems Analyst
- Database Administrator
- Technology Consultant
- Network Administrator
- IT Support Specialist

EOPA

- NOCTI Computer Programming Industry Certification

AP Computer Science Principles

Pathway: Computer Science Pathway

Course Description:

AP Computer Science Principles introduces students to foundational computing concepts through programming, data analysis, cybersecurity, and computational thinking. Students engage in creative problem-solving using Python, JavaScript, and project-based learning experiences. Students investigate computing innovations, analyze data, develop programs, and examine the impact of technology on society while preparing for the AP Computer Science Principles Examination.

Class Grading Scale

- Major Assessments = 55%
- Minor Assessments = 35%
- Practice Assignments = 10%

Career Opportunities

- Software Developer
- Data Scientist
- Cybersecurity Analyst
- Cloud Computing Specialist
- Artificial Intelligence Specialist
- Information Technology Specialist
- Systems Analyst

- Technology Entrepreneur
- UX/UI Designer
- Data Engineer

EOPA

- NOCTI Computer Programming Industry Certification

AP Computer Science A

Pathway: Computer Science Pathway

Course Description:

AP Computer Science A provides students with opportunities to develop object-oriented programming skills through Java programming. Students design, implement, test, and analyze software solutions while working with classes, methods, iteration, selection structures, and data collections. Students prepare for both advanced programming coursework and the AP Computer Science A Examination.

Class Grading Scale

- Major Assessments = 55%
- Minor Assessments = 35%
- Practice Assignments = 10%

Career Opportunities

- Software Engineer
- Java Developer
- Backend Developer
- Application Developer
- Computer Programmer
- Systems Analyst
- DevOps Engineer
- Database Developer
- Solutions Architect
- Software Project Manager

EOPA

- NOCTI Computer Programming Industry Certification

Game Design and Simulation

Pathway: Game Design Pathway

Course Description:

Students develop technical and creative skills through game development experiences using Unity and C#. Students design interactive media, build games, analyze gameplay systems, create user interfaces, and investigate careers within the game development industry while preparing for industry certification opportunities.

Class Grading Scale

- Major Assessments = 55%
- Minor Assessments = 35%
- Practice Assignments = 10%

Career Opportunities

- Game Developer
- Unity Developer
- Gameplay Programmer
- Game Designer
- Level Designer
- Technical Artist
- Interactive Media Designer
- XR/VR Developer
- Simulation Developer
- 3D Artist

EOPA

- Unity Certified Associate: Game Developer

1.2 COURSE PACING GUIDES

IST Semester 1

Georgia Standards

- IT-IST-1 through IT-IST-7

Unit	Timeline
Orientation & Career Exploration	Week 1
Digital Citizenship & Safety	Weeks 2-3
Technology Systems & Troubleshooting	Weeks 4-5
Programming Foundations with Micro:bit	Weeks 6-8
Web Development & UI Design	Weeks 9-10
Robotics & Automation with Finch	Weeks 11-12
Career Exploration & Capstone Project	Weeks 13-18

Major Projects

- Technology Career Research
- Micro:bit Programming Project
- Personal Website Development

FBLA Connections

1. Introduction to Information Technology
2. Website Coding & Development
3. Coding & Programming

CSP Semester 2

Georgia Standards

- IT-CSP-1 through IT-CSP-9

Unit	Timeline
Creative Computing & Digital Artifacts	Weeks 1-2
Programming with JavaScript	Weeks 3-5
Data & Abstraction	Weeks 6-7
Internet & Cybersecurity	Weeks 8-9
Impact of Computing & Ethics	Weeks 10-11
Create Performance Task & Capstone	Weeks 12-18

Major Projects

- JavaScript Application
- Data Visualization Project
- Cybersecurity Investigation
- Computing Ethics Analysis

FBLA Connections

1. Computer Problem Solving
2. Coding & Programming

AP Computer Science Principles

Georgia Standards

- AP CSP Big Idea 1: Creative Development
- AP CSP Big Idea 2: Data
- AP CSP Big Idea 3: Algorithms and Programming
- AP CSP Big Idea 4: Computer Systems and Networks
- AP CSP Big Idea 5: Impact of Computing

Semester 1

Unit	Timeline
Introduction to Computing & Digital Literacy	Weeks 1–2
Programming Foundations with Python	Weeks 3–6
Algorithms & Problem Solving	Weeks 7–9
Data & Information	Weeks 10–12
Internet & Cybersecurity	Weeks 13–15
Impact of Computing & Ethics	Weeks 16–18

Semester 2

Unit	Timeline
Creative Development with Roblox & Lua	Weeks 1–5
Data Science Applications	Weeks 6–8
Advanced Programming Projects	Weeks 9–11
AP Create Performance Task Preparation	Weeks 12–14
AP Create Performance Task Submission	Weeks 15–16
AP Examination Review	Weeks 17–18

Major Projects

- Roblox Interactive Experience
- Data Science Investigation
- Computational Artifact Development
- AP Create Performance Task
- AP Exam Preparation Portfolio

FBLA Connections

1. Computer Problem Solving
2. Coding & Programming
3. Cyber Security

AP Computer Science A

Georgia Standards

- AP Computer Science A Framework

Unit	Timeline
Using Objects and Methods	Weeks 1-7 (semester 1)
Selection and Iteration	Weeks 8-12 (semester 1)
Class Creation	Weeks 13-17 (semester 1)
Data Collections	Weeks 18-30 (semester 2)
AP Examination Review	Weeks 31-36 (semester 2)

Major Projects

- Java Fundamentals Application
- Selection and Iteration Project
- Object-Oriented Programming Project
- Data Collections Application
- AP Practice Portfolio

FBLA Connections

1. Coding & Programming
2. Computer Problem Solving
3. Programming Concepts

Game Design and Simulation

Georgia Standards

- GDAS1 Employability Skills
- GDAS2 Introduction to the Game Industry

- GDAS3 Game Design Process
- GDAS4 Game Asset Development
- GDAS5 Programming and Game Development
- GDAS6 Testing, Debugging, and Deployment
- GDAS7 Career Development and Industry Certification

Semester 1

Unit	Timeline
Introduction to Game Design & Unity	Weeks 1-2
Programming Basics with C#	Weeks 3-5
Game Mechanics & Interactivity	Weeks 6-8
UI Design & Player Feedback	Weeks 9-10
Debugging & Optimization	Weeks 11-12
Unity Game Development Project	Weeks 13-18

Major Projects

- Unity Environment Project
- Interactive Prototype
- Gameplay Systems Project
- User Interface Design Project
- Optimization Analysis
- Original Unity Game

Semester 2

Unit	Timeline
Advanced Unity Development	Weeks 1–3
Asset Creation & Integration	Weeks 4–6
Gameplay Refinement & User Experience	Weeks 7–9

Testing, Debugging, & Optimization	Weeks 10–12
Unity Certification Preparation	Weeks 13–15
Final Game Development Project & Presentation	Weeks 16–18

Major Projects

- Digital Dopamine Game Design Document (GDD)
- Asset Library Development
- Unity Certification Practice Activities
- User Testing Report
- Final Unity Game Project
- End-of-Course Project Presentation

FBLA Connections

1. Game Design
2. Computer Game & Simulation Programming

1.3 FBLA STUDENT ACTIVITIES

IST/CSP

Micro:bit Smart Hall Pass System

FBLA Event: Sales Presentation

Students design and build a Micro:bit-powered hall pass system intended to streamline student movement tracking and improve school operations. Students create a prototype, prepare a product pitch, and present the solution as a technology product.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

AP Computer Science Principles

What's the Odds? Powerball vs. Bitcoin Mining

FBLA Event: Computer Problem Solving

Students investigated and compared the probability of winning the Powerball lottery versus successfully mining Bitcoin. Students researched the problem, gathered supporting information, analyzed the mathematics required for comparison, explored probability concepts, and followed a design-thinking process to develop a proposed software solution before implementation.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

The screenshot shows a presentation slide titled "Powerball research" with a list of bullet points and a mathematical formula. The sidebar on the right contains submission information.

Powerball research

- According to Powerball itself (powerball.com) Each ticket costs 2 dollars
- Doing the math, if you spent the whole 1000\$ on tickets, it would total to 500 tickets
- In order to play powerball, you'd pick 5 numbers from 1-69, as well as a Powerball number from 1-26
- Even though the amount of tickets may sound like we have good odds, the projected jackpot odds are very slim. Doing this math, this is the amount of possible combinations from 5 numbers between 1-69

$$\left(\frac{69}{5} \right) = \frac{69!}{5!(69-5)!} = 11,238,513$$

- Afterwards, we must multiply this by 26, which in total gives us 292,201,338, which means our final odds to win the jackpot prize in the powerball to 1 in 292201338. This is helped by our purchasing of 500 tickets, where it is lessened to 1 in 584,403.
- As well as this, we are almost guaranteed to be able to get more money from smaller prizes, but since it all falls on chance, we ultimately cannot bet on making a profit, let alone making it big

Submission to view: Aug 20 at 12:02 AM

Word Count: 577

Submitted Files: (click to load)
[Bitcoin Vs Powerball by Ismail2.pptx](#)

Assessment

Grade out of 100	Status
99	None

Assignment Comments

Great use of resources and also good analytical skills to identify the right probability model, including math equations to solve the two models before you are able to compare them and generate a solid conclusion of the odds.

AP Computer Science A

Technology Leadership Speaking Challenge

FBLA Event: Impromptu Speaking

Students receive a technology topic, are given twenty minutes to prepare, and deliver a five-to-seven-minute speech demonstrating communication, leadership, and technical understanding.

Evidence Required

- One Graded Student Sample

- Teacher Feedback

Game Design and Simulation

Environment and Emotional Design Analysis

Students evaluate how environmental design, audio, storytelling, color theory, level design, and gameplay mechanics affect player engagement and overall user experience.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

PR-3 Environmental Game Design Review
Due: Aug 28 at 11:59 PM • PR-Game Design Animation and Sim-Torres

12/12 65.42 / 100 (65%) 9/12
Graded Average Students

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Mechanics

The player can move around the place, and interact with the robot citizens there. As you talk to the robots, they give you informations, clues, and even details about their daily life. This allow you to both find clues for your mission and immerse in the environment.

Assignment Comments

i enjoyed it, thanks for sharing. x
Aldrin Torres, Aug 31 at 6:28 PM

As you went through observing and analyzing the gameplay you can realize the importance of the ambient, the environment, every detail counts for capturing the player, to help with the engagement with music, animations, objects, colors that the game intent to allow the player to follow the story through exploration. x

Rubric: **All completed**

- Explain location
- use of elements
- Mechanics
- Level of Intensity
- How to spark Intensity
- what creates emotions
- thoughts about the overall story

1.5 LONG-TERM PROJECTS

Program Theme

Digital Wellness, Human Behavior, and Technology

Computer Science Pathway

Digital Wellness Analytics Platform

Community Partner: Neuroscience & Psychology Club

Students develop a web-based analytics platform that investigates the relationship between technology usage habits and student wellness. Students create anonymous survey instruments, collect voluntary anonymous survey data, organize and analyze datasets, develop visualizations, and publish findings through an interactive website dashboard.

Deliverables

- Project Proposal
- Anonymous Survey Instrument
- Dataset
- Data Analysis Report
- Interactive Website Dashboard
- Final Presentation

Total Time

23 Hours

Industry Skills

- Data Collection
- Data Cleaning
- Data Visualization
- Front-End Web Development
- User Experience Design
- Software Testing
- Project Documentation

Game Design Pathway

Digital Dopamine: Understanding Technology Through Interactive Play

Community Partner: Neuroscience & Psychology Club

Students develop a lightweight 2D Unity educational game that demonstrates how technology habits influence focus, attention, reward systems, habit formation, and digital wellness. The game emphasizes simple assets, scene transitions, educational gameplay mechanics, and accessibility within school hardware constraints.

Deliverables

- Game Design Document
- Storyboard
- Asset Library
- Playable Unity Prototype
- Testing Report
- Final Presentation

Total Time

22 Hours

Industry Skills

- Game Design
- Unity Development
- C# Programming
- User Interface Design
- Debugging
- Asset Management
- Interactive Storytelling